

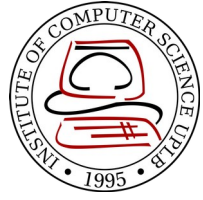
# *CMSC 137*

## *Data Communications and Networking*

Lecture Module  
04 – OSI Model



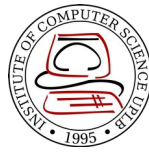
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# Module 4: OSI Reference Model

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# Recall

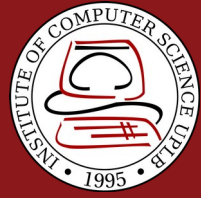


- Network is a combination of hardware and software that sends data from one location to another
- Hardware consists of physical equipment that carries signals
- Software consists of instruction sets that make possible the services that we expect from a network
- There are different companies that produce networking equipment

# OSI Reference Model

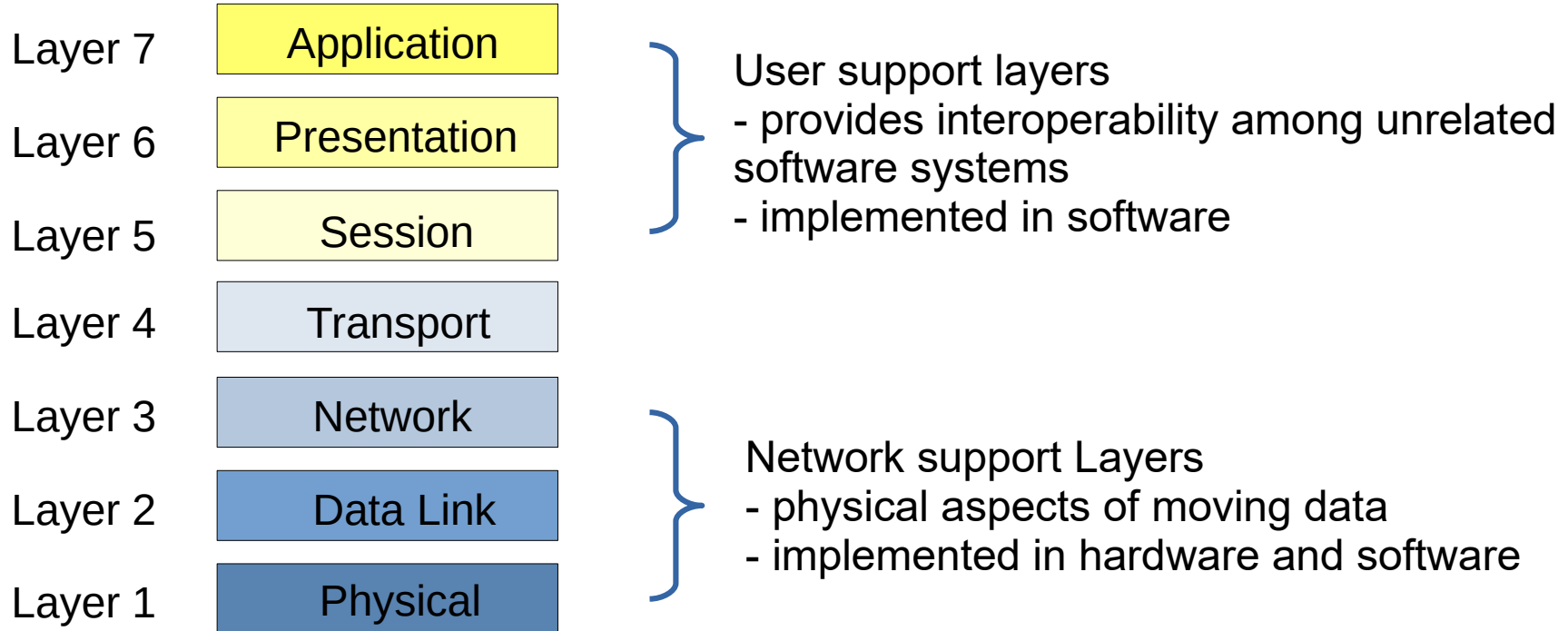
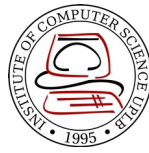


- International Standards Organization (ISO) introduced the Open Systems Interconnection model in the late 1970s
- [OSI model ISO/IEC 7498-1](#)
- [Open system](#) – a set of protocols that allows any two different systems to communicate regardless of the underlying architecture
- [OSI is not a protocol](#); it is model for understanding and designing a network architecture that is flexible, robust, and interoperable
- Layered framework for the design of network systems that allow communication between all types of computer systems



# 7 Layers

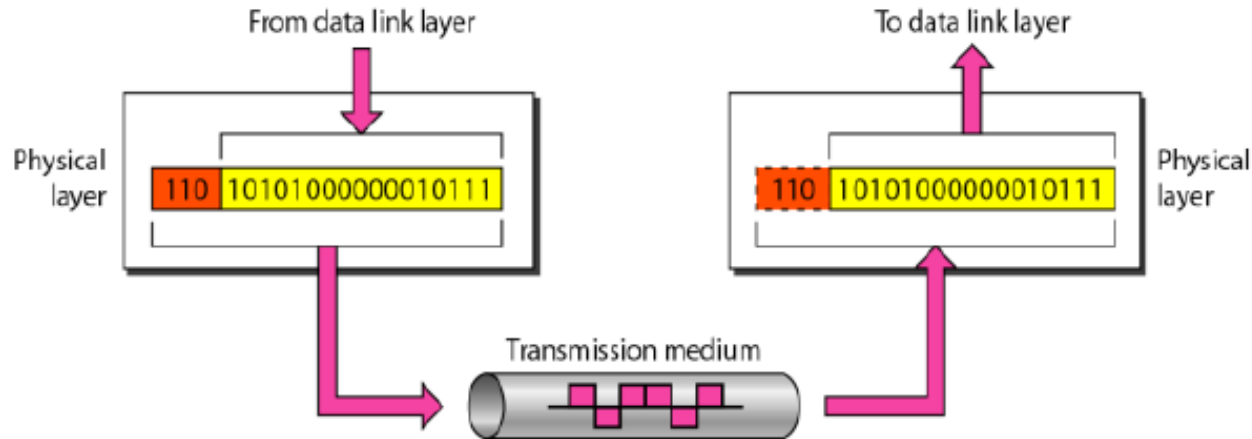
# 7 Layers of the OSI Reference Model



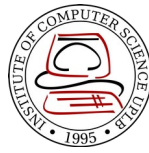
# Physical Layer



- Defines the connector and interface specifications as well as the medium (cable) requirements
- Electrical, mechanical, functional and procedural specifications are provided for sending a **bit stream** on a computer network



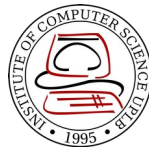
# Physical Layer



- Physical characteristics of interfaces and medium
- Representation of bits - encoding
- Data rate - number of bits sent each second
- Synchronization of bits - timing
- Line configuration – point-to-point, multipoint
- Physical topology – bus, star, ring, mesh, hybrid
- Transmission mode – simplex, half-duplex, duplex



# Some components at Layer 1

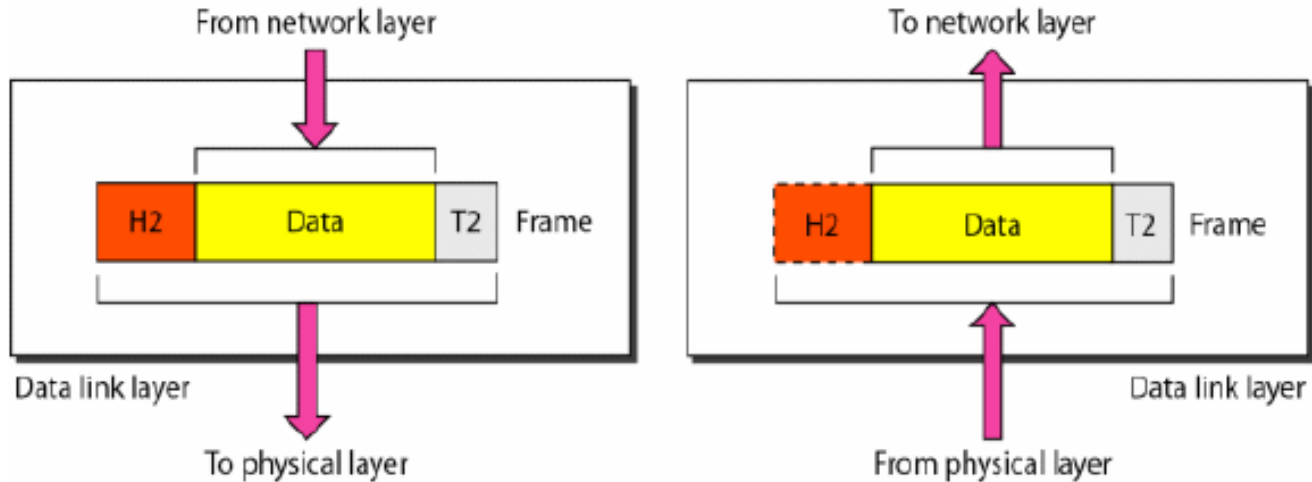


- Cabling system components
- Adapters that connect media to physical interfaces
- Connector design and pin assignments
- Hub, repeater and patch panel specifications
- Wireless system components
- Parallel SCSI
- Network Interface Card (NIC)

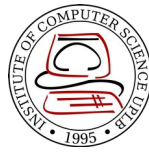
# Data Link Layer



- Transforms the physical layer to a reliable link
- Makes the physical layer appear error-free to the upper layer



# Data Link Layer



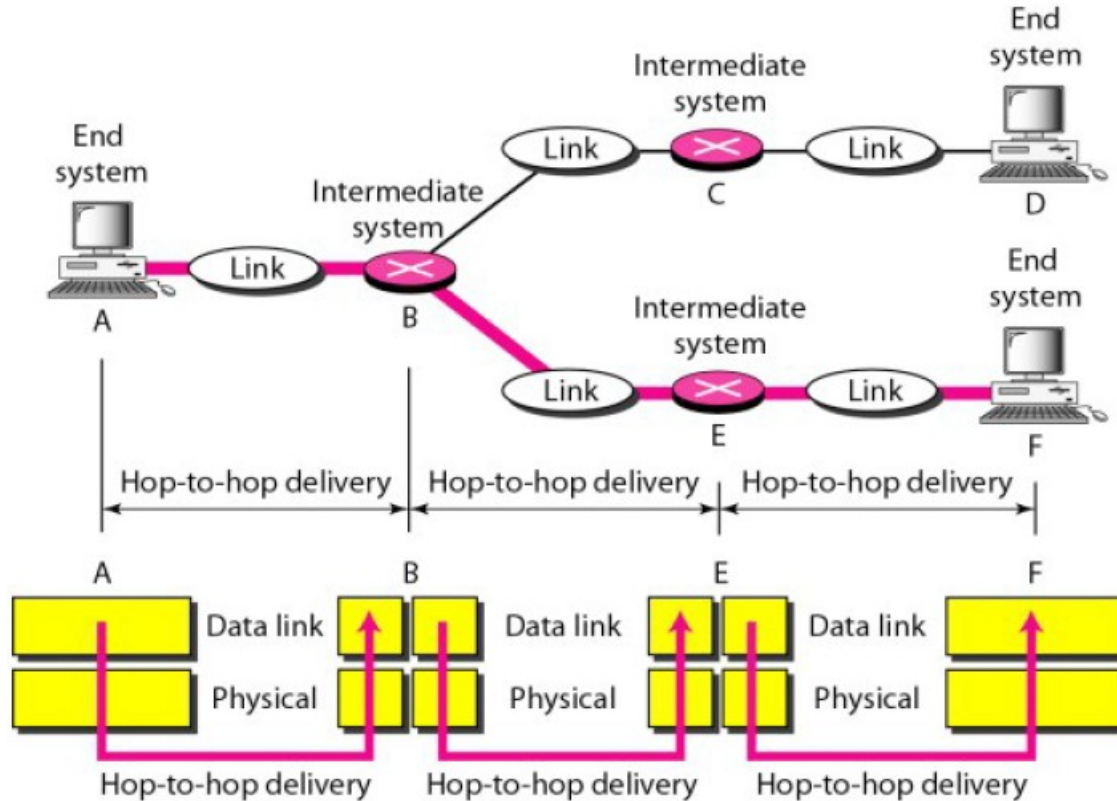
- Framing – defining frames
- Data link addressing – defines sender/receiver
- Flow control – prevents overwhelming the receiver
- Error control – detect and re-transmit damaged or lost frames, use of trailers
- Access control – defines which device has control over the link when it is shared by devices

# Some components at Layer 2



- Ethernet and Token Ring switches
- Bridges

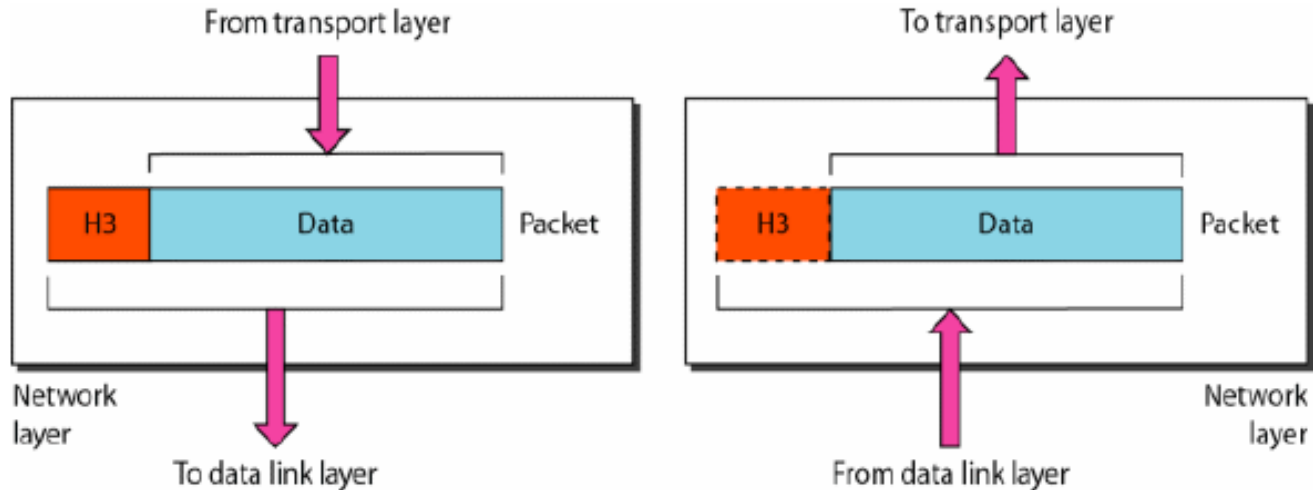
# Hop to hop delivery



# Network Layer



- **Source-to-destination** delivery of packets, possibly across multiple networks
- No need for network layer if two systems are connected to the same link

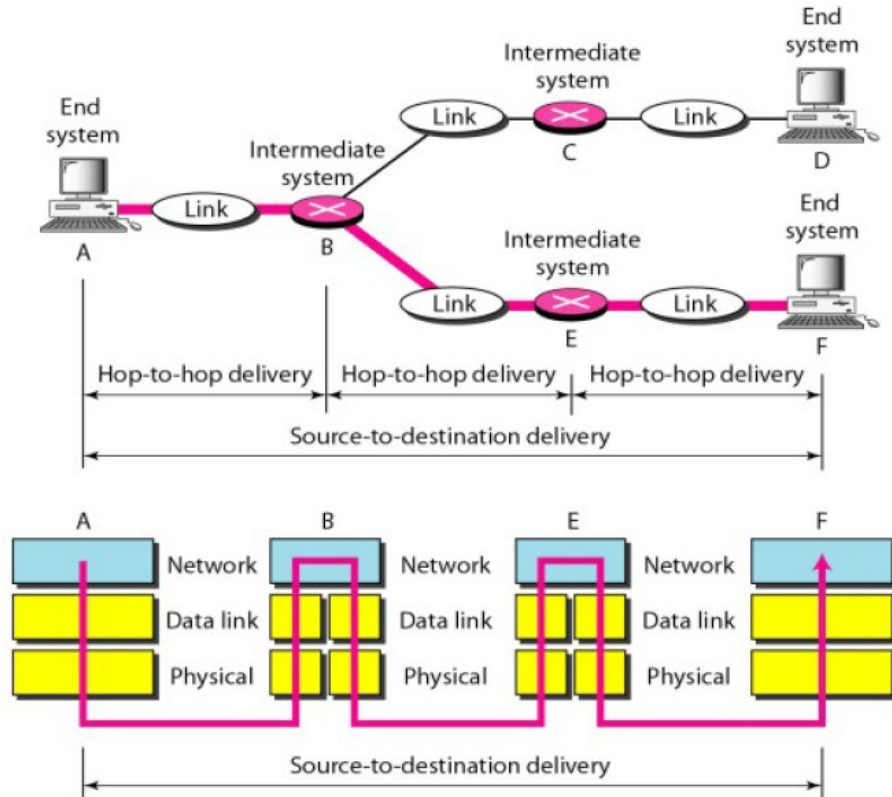


# Network Layer



- **Logical addressing** – distinguish source and destination systems if packets pass network boundary
- **Routing** – makes possible the delivery of packets in internetworks, connecting devices are called routers and switches

# Source to destination delivery

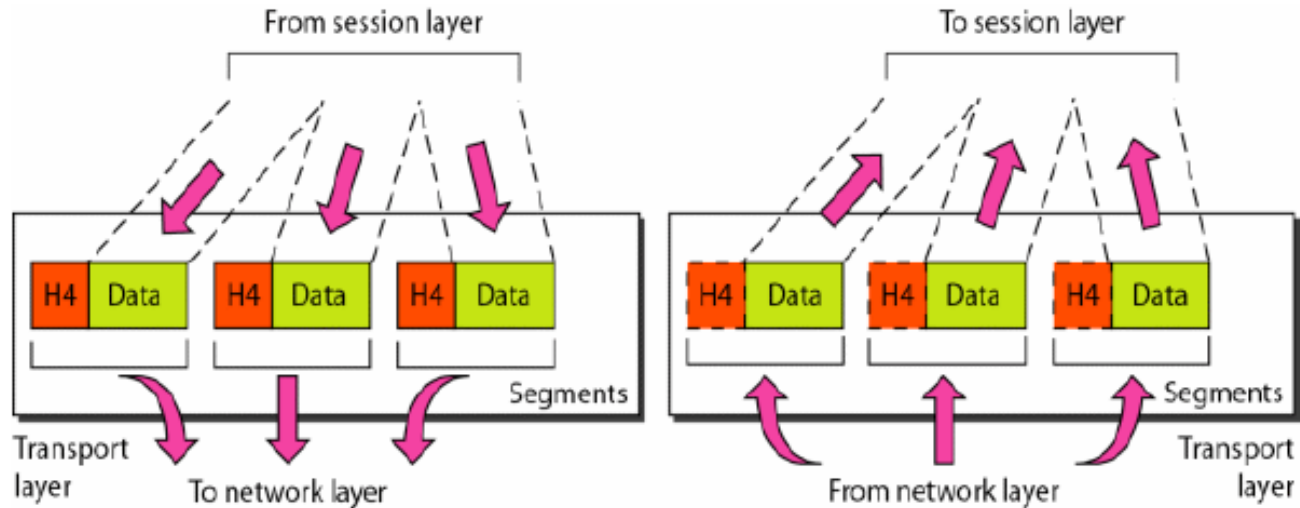




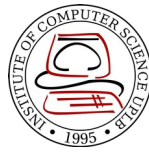
# Transport Layer



- Responsible for **process-to-process** delivery
- Ensures that the whole message arrives intact and in order

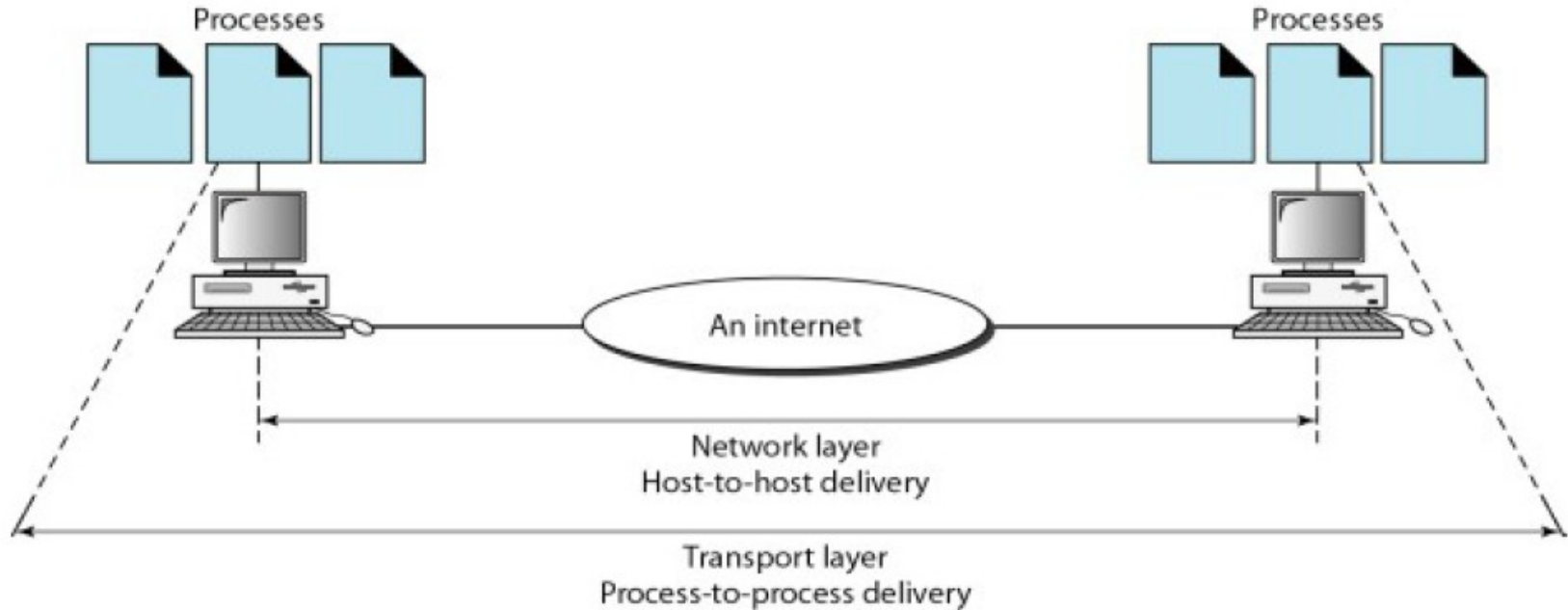
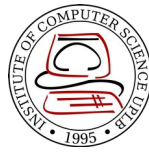


# Transport Layer



- **Service-point addressing** – several programs (entities) can use the network, uses ports
- **Segmentation and reassembly** – message divided into transmittable segments (uses sequence numbers)
- Connection control – maybe connectionless or connection-oriented
- Flow control – end-to-end flow control rather than a single link
- Error control – process-to-process flow control rather than single link

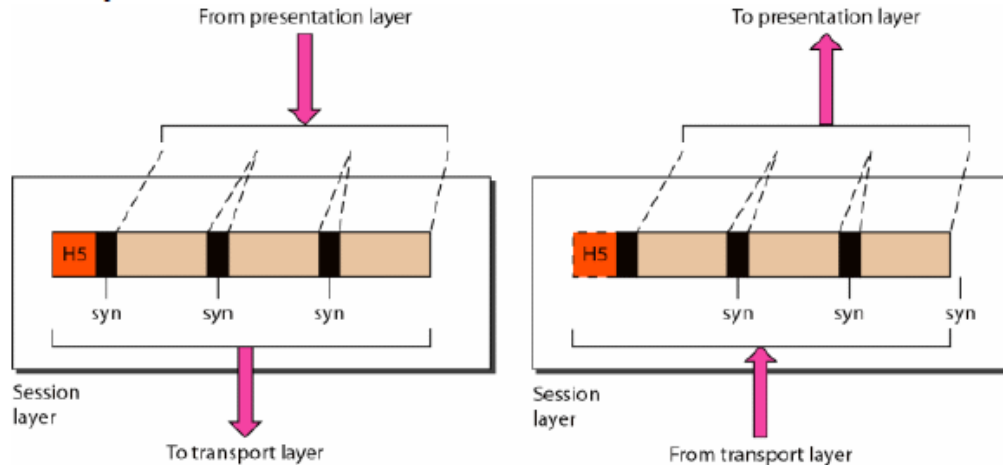
# Process to process delivery



# Session Layer



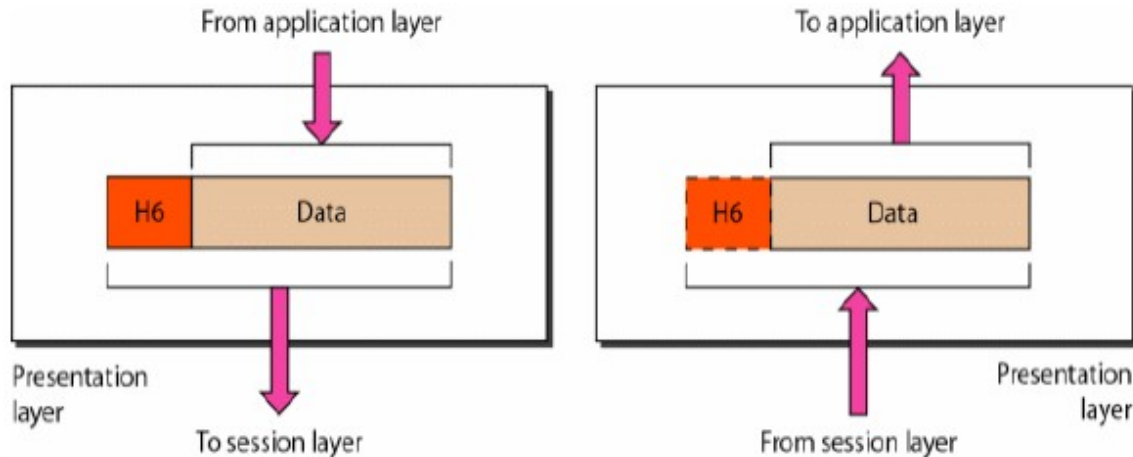
- Network dialog controller
- Establishes, maintains, and synchronizes interactions among communicating systems
- Dialog control – dialog between two systems
- Synchronization – add checkpoints (similar to databases)



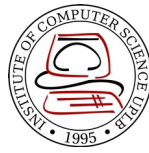
# Presentation Layer



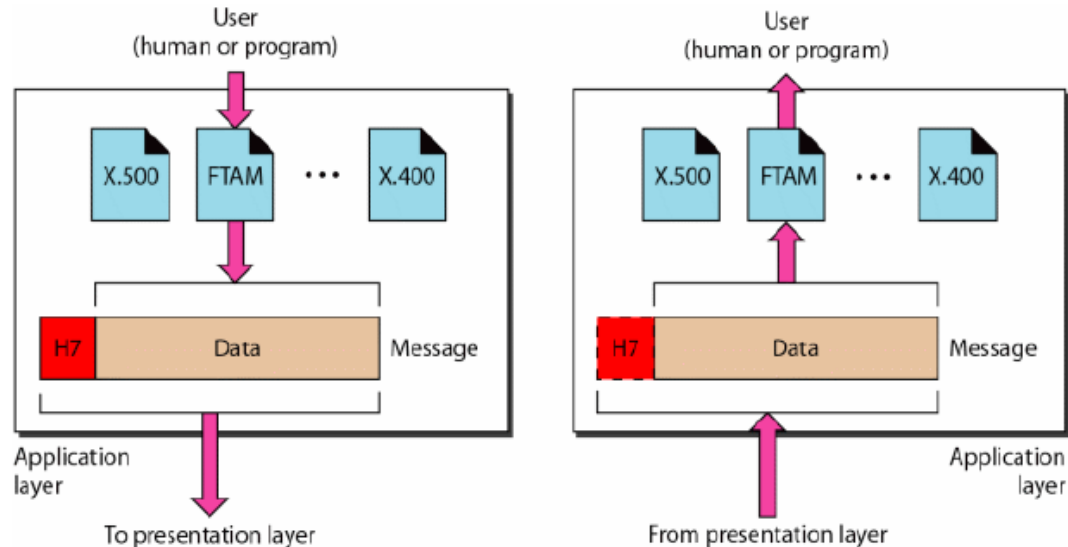
- Concerned with the syntax and semantics of the information exchanged between two systems
- **Translation** – apply encoding schemes
- **Encryption** – provides privacy
- **Compression** – reduce the number of bits contained in the information



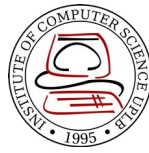
# Application Layer



- Enables the user (human or software) to access the network
- Provides user interfaces and support services: email, remote file access and transfer, shared DBM, distributed information services

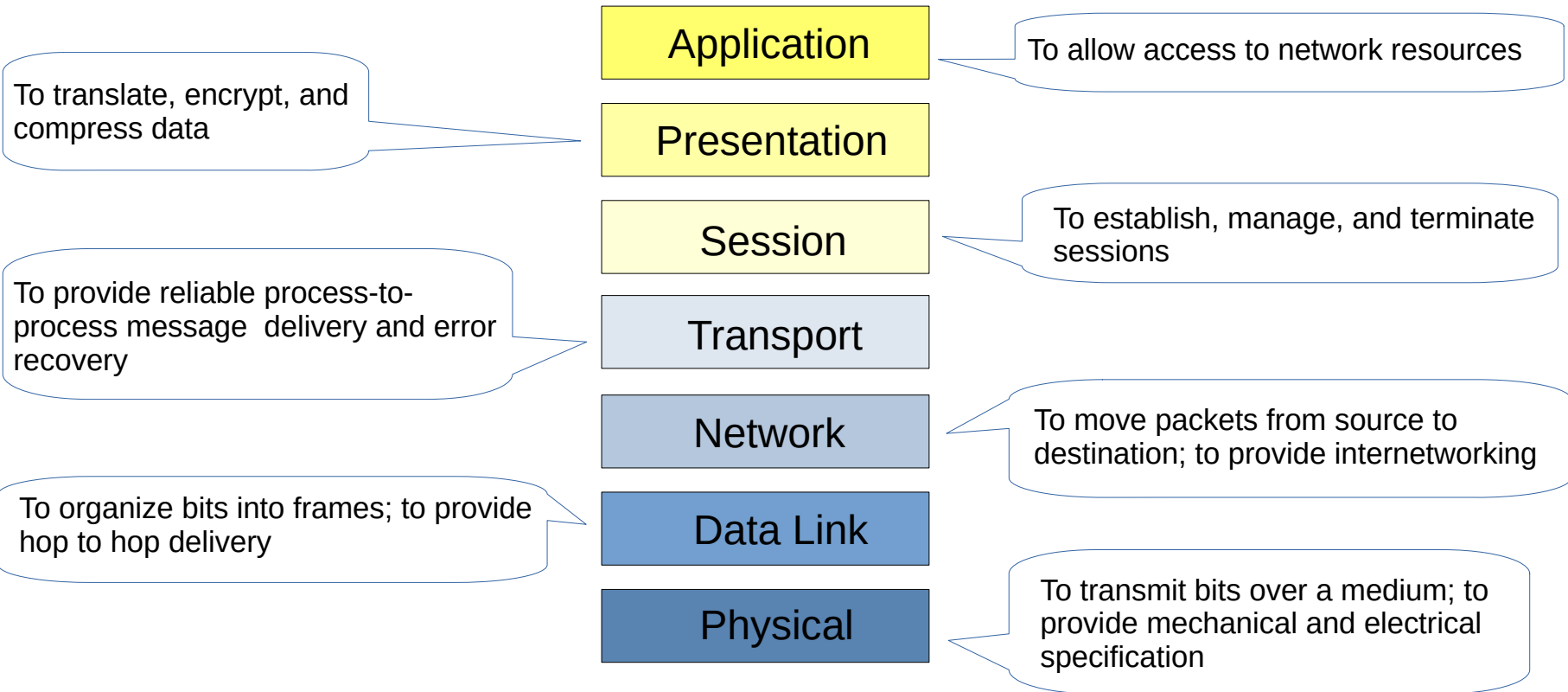
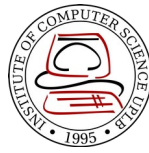


# Application Layer



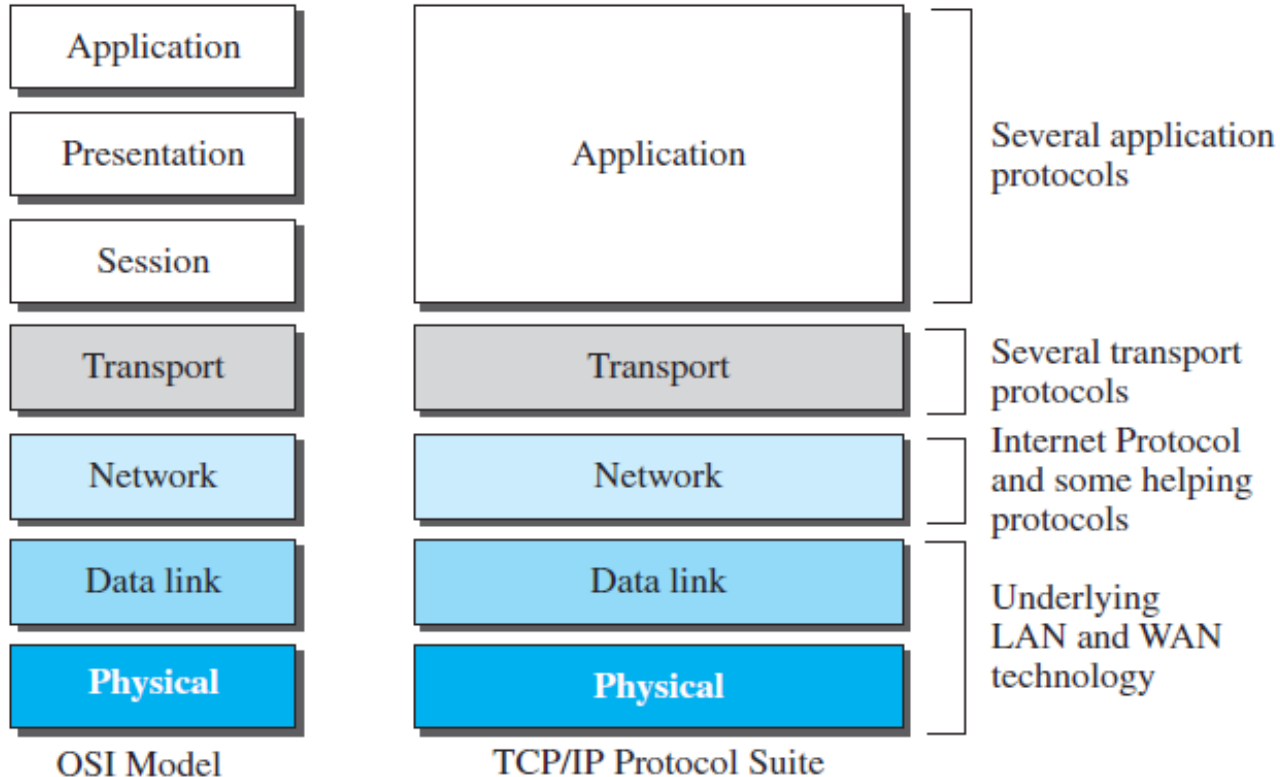
- Network virtual terminal – software version of a software terminal
- File transfer, access, and management – access files remotely
- Mail services – email forwarding and storage
- Directory services – distributed database sources

# OSI Model Summary





# OSI vs TCP/IP



# Lack of OSI Model's success



- TCP/IP was fully in place; changing it would cost a lot
- Some layers are not fully defined
  - Presentation and session layers actual protocols were not fully defined, nor fully described and corresponding software was not fully developed
- When OSI was implemented by an organization in a different application, it did not show high enough level of performance to entice the Internet authority to switch

# Lack of OSI Model's success



Read “The Internet that wasn’t” by Andrew L. Russell

- <https://spectrum.ieee.org/tech-history/cyberspace/osi-the-internet-that-wasnt>
- <https://ieeexplore.ieee.org/document/6565559>

# Reference



- Forouzan, B.A. 2013. Data Communications and Networking, 5th Ed. McGraw-Hill, New York.